CY33/CI33

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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – FEB / MARCH 2025

Program	:	**B.E. – CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)**	**Semester**	:	**III**
Course Name	:	**Data Structures**	**Max. Marks**	:	**100**
Course Code	:	**CY33 / CI33 (2021,2022,2023 Batch)**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	Differentiate between NULL and generic pointer with suitable examples.	CO1	(06)
	b)	What is the role of free() and realloc() functions in Dynamic memory allocation. Explain with suitable C code snippets.	CO1	(06)
	c)	Explain Linear and Nonlinear Data Structure categories with examples.	CO1	(08)
2.	a)	Write a C program to swap two numbers using pass by value and pass by reference.	CO1	(06)
	b)	Explain calloc() and malloc() Dynamic memory allocation functions with suitable C code snippets.	CO1	(08)
	c)	Discus the different operations on Data Structures.	CO1	(06)

UNIT - II

3.	a)	Convert the following infix expression to postfix form using stack. ((M+N)∗O−P)/(Q+R∗S)	CO2	(08)
	b)	Evaluate a given postfix expression using stack. 23∧ 56+* 83+/	CO2	(04)
	c)	Write algorithm to insert and delete element in Circular Queue.	CO2	(08)
4.	a)	Convert the following infix expression into prefix form using stack. (A – B / C) * (A / K – L).	CO2	(10)
	b)	Write a C program to implement input restricted Dequeue.	CO2	(06)
	c)	Write a C program to print Fibonacci series up to 10 numbers using recursion.	CO2	(04)

UNIT - III

5.	a)	Explain linear queue? What are the applications of linear queue? Write a C program to simulate the i) insert ii) delete iii) display operations on linear queue.	CO3	(06)
	b)	Explain how would you implement a circular queue using dynamically allocated arrays.	CO3	(07)
	c)	Explain circular queue? What ate the advantages of Circular queue over simple queue.	CO3	(07)
6.	a)	Illustrate multiple queues and its advantages.	CO3	(06)
	b)	Explain priority queues and its usage.	CO3	(07)
	c)	Explain the applications of Queues.	CO3	(07)

UNIT- IV

7. a) Write Algorithm for in-order traversal of a Threaded Binary tree. CO4 (08)

b) Construct a Binary tree by using the following traversals. CO4 (12)
In–order Traversal: D B H E I A F J C G
Post order Traversal: D H I E B J F G C A

8. a) Discuss the importance of balance factor in AVL trees. CO4 (05)

b) Perform the inorder, preorder and postorder traversals for the given Binary tree in Fig.8(b). CO4 (10)

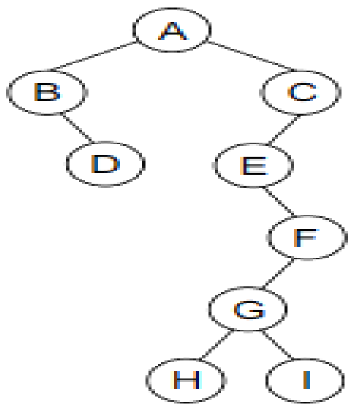

Fig.8(b) Binary Tree

c) Consider the given the binary tree in Fig. 8(c), Write down the expression that it represents. CO4 (05)

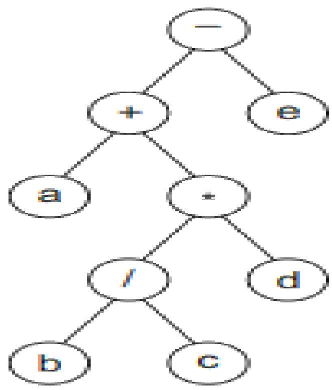

Fig. 8(c) Binary Tree

UNIT – V

9. a) What is collision in Hashing? Explain different collision resolution techniques with suitable examples. CO5 (12)

b) Create a B tree of order 5 by inserting the following elements: CO5 (08)
3, 14, 7, 1, 8, 5, 11, 17, 13, 6, 23, 12, 20, 26, 4, 16, 18, 24, 25, and 19.

10. a) Write Algorithm for inserting a new node and deleting a node in a B+ tree. CO5 (10)

b) Consider a Hash table of size 10. Using Quadratic probing, insert the keys 72, 27, 36, 24, 63, 81, and 101 into the table. Take c1 = 1 and c2 = 3. CO5 (10)

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